جامعة خليفة
Khalifa University

Introduction to Software Engineering
COSC 336
ReSource (Phase II)

Feasibility Study

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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Introduction

Concept

ReSource is a proposed Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System for Khalifa University. It would help departments identify, share, and reuse surplus or underutilized assets, such as furniture, computers, laboratory equipment, vehicles, and books. A department requesting an asset could check available campus resources before purchasing a new item.

The system would track each asset throughout its life cycle, including registration, allocation, inspection, maintenance, repair, transfer, and eventual donation, recycling, or disposal. It combines traditional asset management with AI-assisted classification, semantic matching, sustainability recommendations, natural-language interaction, and generated reports. Authorized users would remain responsible for approving significant asset decisions [1].

Document Purpose

This document presents the Phase II feasibility study for ReSource. It builds on the scope, stakeholders, and initial requirements identified in Phase I to assess whether the proposed system can be developed and demonstrated within the available time and resources.

The study examines market demand and existing alternatives, technical requirements, financial viability, operational adoption, and schedule feasibility. It also considers data availability, AI suitability, and relevant legal, ethical, privacy, security, ownership, procurement, and safety constraints. Its purpose is to support a decision on whether to proceed and identify the conditions and risks that should guide the following phases [1], [2].

Project Overview

High-Level System Description

ReSource is proposed as a web-based system built on two connected layers. The classic layer manages asset records and controlled workflows, while the AI-powered layer provides recommendations and assistance using information held in the system.

The following high-level functional requirements are derived from the project instructions [1]. Each uses a unique FR identifier, meaning Functional Requirement, to support traceability to stakeholder needs, design components, and test cases. Detailed requirements and acceptance criteria will be developed during Phase III.

Layer 1 Classic Asset Management

Module	Function
FR-1 Asset Registration	The system shall allow authorized users to register and update asset descriptions, categories, quantities, departments, ownership, locations, condition, availability, estimated values, and supporting information.
FR-2 Resource Marketplace	The system shall publish available surplus assets and allow authorized users to search and filter listings using relevant asset attributes.
FR-3 Request & Reservation	The system shall allow departments to describe resource needs, submit requests, reserve available assets, and track request status.
FR-4 Approval & Transfer	The system shall record authorized approval decisions, transfer arrangements, receipt confirmation, and changes in ownership, custody, and location.
FR-5 Life-Cycle Tracking	The system shall maintain histories covering inspections, maintenance, repairs, transfers, refurbishment, donation, recycling, retirement, and disposal.
FR-6 Access Management	The system shall authenticate users and restrict access to records and actions according to assigned roles and departmental responsibilities.
FR-7 Operational Reporting	The system shall provide status notifications, audit records, and conventional reports on inventory, requests, approvals, transfers, maintenance, and asset status.

Layer 2 AI-Powered Features

Module	Function
FR-8 AI Matching Engine	The system shall rank suitable assets against requests using semantic similarity and relevant specifications, quantity, condition, location, urgency, dates, and available costs, with an explanation for each recommendation.
FR-9 AI Classification	The system shall suggest categories, tags, and standardized descriptions that authorized users can review and correct.

FR-10 Sustainability Optimizer	The system shall recommend continued use, transfer, repair, refurbishment, donation, component reuse, recycling, or disposal using available asset information and stated decision criteria.
FR-11 LLM Assistant	The system shall support natural-language searches, asset questions, request preparation, status tracking, and navigation, while respecting permissions and obtaining confirmation before submitting or changing records.
FR-12 Generative Reporting	The system shall generate summaries from recorded operational and sustainability information, identifying assumptions and missing data behind estimates.

The classic layer shall enforce permissions, availability, quantities, and approvals. AI recommendations shall remain subject to authorized human review. Structured search, ordinary forms, and conventional reports shall remain available when AI services fail or produce unsuitable results.

Background and Context

The course project brief identifies a campus resource-management problem: departments may purchase new resources while suitable items remain unused elsewhere. It also explains that traditional asset registers may record ownership and location without providing the internal exchange, matching, and sustainability functions needed to support reuse [1]. This establishes the problem being investigated for ReSource.

There is external evidence that universities use shared platforms to redistribute surplus resources. For example, Warp It's published University College London case study describes an institutional reuse platform [3]. This supports the relevance of campus resource exchange, but it does not establish Khalifa University's current procedures or expected savings.

The current process at Khalifa University has not yet been verified through real stakeholder interviews. Statements about spreadsheets, email, paper approvals, or informal disposal arrangements are therefore working assumptions. The AI-simulated interviews in Appendix A of the full report are planning exercises and do not count as statements from university employees.

ReSource would provide a shared inventory, an internal marketplace, controlled transfers, and life-cycle records. Stakeholder interviews are still needed to establish which existing processes it would support or change, what information is available, and who holds approval authority.

Business Goals and Objectives

- Reduce unnecessary purchases by identifying suitable surplus assets before buying new resources.

- Extend asset life through maintenance, repair, refurbishment, and accurate life-cycle records.
- Increase appropriate reuse, donation, and recycling while maintaining human approval over asset decisions.
- Provide traceable information about avoided purchases, savings, waste diversion, and other supported sustainability indicators.
- Compare classic search, reporting, and decision rules with AI-enhanced approaches to evaluate their benefits and limitations.

Expected Benefits

- Financial: Potential reductions in unnecessary purchases and storage or disposal costs where reuse is appropriate.
- Operational: Clearer request tracking, approval responsibilities, transfer confirmation, and maintenance records.
- Environmental: Greater opportunities to keep usable assets in service and divert suitable materials from disposal.
- Organizational: A shared source of information about asset availability, location, condition, custody, and history.

These are expected outcomes. Their scale would depend on participation, data quality, transfer arrangements, and operating costs. Demonstrations using synthetic data would not establish actual university savings or environmental improvements.

Assumptions

- University stakeholders would consider an internal exchange platform if it fits their responsibilities and approval procedures.
- Departments would assign staff to register surplus assets and maintain accurate availability and condition information.
- Intended users would have browser access and sufficient computer skills to use the system with basic guidance.
- Relevant stakeholders would be available to review workflows, permissions, and evaluation criteria.

These assumptions require validation during stakeholder consultation and prototype evaluation.

Project Constraints

- Timeline: Development must fit within the Fall 2026 semester and the confirmed deadlines for the eight project phases.
- Team size: Four students will share requirements analysis, design, implementation, testing, and documentation.
- Prototype scope: The course project will demonstrate workflows and AI functions at a manageable scale. University-wide deployment would require additional work.

- Technical resources: AI functions will use existing libraries, pretrained models, or services. Custom training of large models is outside the available resources.
- Data availability: Real university asset data is unavailable for the course project. Development and evaluation will use reviewed sample or synthetic records, with their limitations documented.

Project Scope

The feasibility study assesses the following proposed scope. Implementation priorities and detailed acceptance criteria will be refined during Phase III using the feasibility findings.

In scope	**Out of scope**
Asset registration, surplus publication, departmental requests, structured search, reservations, approvals, transfers, maintenance, life-cycle history, and role-based access.	Live integration with university procurement, finance, or other institutional systems during the academic prototype.
AI classification, semantic matching and ranking with explanations, sustainability recommendations, natural-language assistance, and generated summaries at prototype scale.	Large-scale custom model training or continuously retrained AI infrastructure.
Notifications, audit records, conventional reports, and sustainability indicators supported by available data and stated assumptions.	University-wide production deployment, formal certification or compliance auditing, and verified campus savings claims based on synthetic data.

Development would prioritize a complete classic workflow and core AI demonstrations before expanding supporting features. Any reduction in AI scope would need to remain consistent with the project instructions and be confirmed with the lab instructors.

AI recommendations would not independently authorize transfers, purchases, repairs, donations, recycling, or disposal. These actions would remain subject to relevant human approval and safety procedures.

Stakeholders

ReSource would affect staff who register assets, request resources, approve transfers, maintain equipment, administer the system, and review financial or sustainability outcomes. The project instructions identify these roles as relevant to the system and its feasibility assessment [1].

The table defines proposed responsibilities and permissions for validation. It does not establish university appointments or existing approval procedures.

Stakeholder (Category)	Role/Responsibility	Needs/Expectations & Data Needs	Feature Requirement Mapping	Permissions & Approval Decisions
Administrators (Business User)	Coordinate requests, resolve conflicts, and review transfers and end-of-life actions.	Approval queues, asset status, transfer details, and audit history.	FR-4, FR-7, FR-10	Approve business actions within assigned authority.
Procurement Team (User)	Consider reuse before purchasing and review purchase justification.	Inventory, specifications, costs, and avoided-purchase evidence.	FR-2, FR-3, FR-7, FR-8, FR-12	Review reuse and purchase recommendations within procurement authority.
Department Representatives (User)	Identify needs and coordinate surplus listings and requests.	Availability, suitable matches, and request status.	FR-2, FR-3, FR-8, FR-11	Manage authorized departmental listings and submit requests.
Asset Coordinators (User)	Maintain asset condition, availability, custody, and location.	Accurate asset histories, update forms, and transfer information.	FR-1, FR-4, FR-5, FR-9	Update assigned assets and confirm receipt and condition.
Maintenance Staff (User)	Inspect assets and record defects, repairs, costs, and outcomes.	Specifications, condition, and maintenance history.	FR-5, FR-10	Update maintenance records and recommend service actions.
Finance Officers (User)	Review values, costs, and savings calculations.	Financial records, calculation inputs, and supporting evidence.	FR-7, FR-12	Access authorized financial data and review assumptions.
Sustainability Officers (User)	Review reuse outcomes and environmental indicators.	Reuse records, methods, and stated uncertainties.	FR-7, FR-10, FR-12	Review sustainability information and recommend actions.

System Administrators (Technical User)	Configure accounts, roles, services, backups, and monitoring.	Account records, configuration, service status, and technical logs.	FR-6, FR-7	Administer technical access. No automatic asset-transfer or disposal authority.
University Management (Customer)	Confirm priorities, resources, and adoption decisions.	Feasibility findings, costs, risks, and management reports.	FR-7, FR-12	Authorize sponsorship, budget, and adoption through university procedures.
IT Department (Technical Owner)	Oversee hosting, security, maintenance, and support.	Architecture, service dependencies, and support documentation.	FR-6, FR-7 and infrastructure	Oversee technical arrangements and system administration.
Developers (Software Developer)	Analyze, design, implement, test, and document the prototype.	Agreed requirements, tools, test data, and feedback.	FR-1 to FR-12	Manage development without university business approval authority.
Lab Instructors (Development Manager)	Guide the project and assess deliverables.	Documentation, progress evidence, and evaluation results.	Project process and assessment	Review and assess course deliverables.

The proposed business owner is the Head of Facilities Management, Khalifa University, responsible for the system's operational purpose and proposed adoption. The proposed technical owner is the IT Department. The individual representatives and formal responsibilities require confirmation. During the academic project, the student team would manage the development and demonstration environment.

Stakeholder Validation

Real interviews are proposed before completing Phase III. Ahmed AlAbdouli is the proposed interview coordinator, subject to team confirmation. The interviews remain pending and would address the following questions:

- Facilities Management and asset coordinators: How are surplus assets identified and recorded, and who approves release, transfer, and receipt?
- Procurement and finance: When must available inventory be checked, and what evidence supports an avoided purchase or savings calculation?
- Maintenance and sustainability staff: Which inspections, safety checks, repair criteria, and environmental calculation methods are required?
- IT and system administrators: Which hosting, access, backup, data-handling, and external AI-service restrictions would apply?

- Department representatives: Which resources are needed or held in surplus, who would maintain listings, and what would encourage participation?

For each completed interview, the team should record the participant's name and role, date, interviewer, confirmed findings, and unresolved questions, with permission to include the information in the report. Participant statements should remain distinguishable from the team's interpretation.

Market Analysis

Target Market and Demand

ReSource's immediate target market is Khalifa University's internal asset-management environment. Facilities Management and university management would be the principal organizational customers, while departments and operational staff would use the system to identify, request, transfer, and maintain resources.

The Phase II slides require an assessment of demand and competition [2]. For ReSource, demand should be assessed through the availability of reusable assets, departmental participation, and useful exchanges. Public sales or subscription revenue are not the primary measures for this internal academic prototype.

The project brief establishes the campus reuse problem [1], while the published University College London case study provides an example of institutional adoption elsewhere [3]. This suggests that the service category is relevant, but local demand remains unverified. No confirmed campus inventory analysis or real stakeholder interview results are available for this study.

Existing Solutions and Competition

The comparison uses official product descriptions. The implications for ReSource are preliminary assessments, not findings from hands-on product testing.

Existing Solution or Alternative	**Relevant Capabilities**	**Implications for ReSource**
Warp It	Surplus listings, resource discovery, reuse transactions, and reporting on avoided costs and waste diversion. Its website also describes AI-assisted listing information [4].	A direct comparison for campus reuse. ReSource should demonstrate workflow suitability rather than claim that reuse or AI assistance is unique.
Rheaply	An internal reuse marketplace with tracking of cost savings, waste diversion, and embodied carbon avoidance [5].	Shows that internal exchange and sustainability reporting are established capabilities. A future deployment should compare operational fit and support needs.

Snipe-IT	Asset records, assignment, check-in/check-out, and role-based permissions [6].	A baseline for inventory and custody management. Suitability for the full campus exchange and AI workflow needs further assessment.
Spreadsheets and email	Could support a basic register and communication, with manual coordination of updates and approvals.	A simple alternative to assess during interviews. Their actual use at Khalifa University has not been verified.

Proposed Value of ReSource

ReSource would combine resource exchange, controlled transfers, maintenance history, and AI assistance around the project's campus scenarios. Its proposed value would depend on whether this combination helps users complete tasks more effectively than the alternatives.

The team would evaluate semantic matching against structured search, AI classification against reviewed categories, and generated summaries against conventional reports. Clear approval boundaries and traceable financial and environmental estimates would also form part of the evaluation.

The academic purpose supports developing a prototype to investigate these functions. A future deployment would require a separate comparison of building ReSource, configuring an existing product, or improving current processes.

Adoption Factors and Demand Validation

Adoption would depend on accurate listings, clear responsibilities, suitable approval procedures, and practical arrangements for inspection and collection or delivery. The proposed stakeholder interviews should establish:

- Which assets departments need or hold in surplus and how frequently suitable exchanges could occur.
- Who would maintain records and confirm availability.
- Whether the proposed request and approval workflow fits departmental responsibilities.
- Whether AI suggestions provide useful assistance compared with structured search.

Where access is permitted, the team should support interview findings with anonymized examples of asset registers, request forms, maintenance records, or transfer procedures. Synthetic data can demonstrate system behavior, but it cannot establish actual campus demand or financial returns.

Market Analysis Conclusion

The preliminary analysis identifies a plausible campus reuse need and existing products that address comparable functions. This supports continued prototype evaluation. An operational adoption decision would require verified stakeholder demand, assessment of existing

alternatives, and evidence that expected benefits justify implementation and ongoing support costs.

Technical Analysis

1) Hardware and Software Requirements

ReSource is proposed as a web-based system that combines campus asset management with AI-assisted resource matching, classification, and sustainability recommendations. Development will use free and open-source tools, pretrained models, or APIs, consistent with the scope defined in Phase 1.

- **Backend (Python/Django or Java)**: The backend would process asset registrations, requests, reservations, approvals, transfers, and maintenance records. Django provides user authentication of accounts, groups, permissions, and password hashing [7].
- **Frontend (HTML, CSS, and JavaScript)**: The interface would provide asset listings, search filters, registration forms, request tracking, and dashboards appropriate to each user's role.
- **Database (MySQL)** : Database would store users, departments, assets, requests, transfers, and life-cycle records. Linking these records would support ownership tracking and reporting. MySQL is supported by Django as well.
- **AI Libraries**: Use pre-defined AI libraries such as (Sentence Transformers) for semantic search, comparing asset and description. LLM could also be used to support generated reports and natural-language interaction
- **Hardware**: Existing teams' and labs computers would be used for development and testing.

For example, when a department needs equipment, a user would enter the request through the frontend (HTML, CSS, and JavaScript). The backend (Python/Django or Java) would process the request and retrieve available asset records from the MySQL database. Sentence Transformers would compare the request's meaning with asset descriptions to help identify suitable matches. The backend would apply checks for availability and required specifications, and the frontend would display the results.

When the user selects an asset, the backend would verify permissions and quantities, record the reservation in MySQL, and route the request for approval. After the transfer is confirmed, it would update the asset's custody, location, and life-cycle history in the database. An LLM could subsequently generate a summary using the recorded transfer information.

2) Skills Required

The project requires the following steps:

- **Software Engineering skills**: Designing/Planning, Implementing, and testing skills
- **Web Development**: UI, and backend workflows. Making the system easy to use.
- **Database Design**: Connecting assets, user, request, and maintenance records while maintaining accurate quantities and ownership.
- **AI integration** (layer 2 of the system): Using pretrained models or APIs, preparing input data, evaluating recommendations, and handling unsuccessful responses.
- **Collaboration and deployment:** Managing shared code through GitHub, integrating modules, and configuring the demonstration environment.

For example: One team member develops the interface, another builds the backend and approval workflows, a third designs the database for asset and transfer records, and the fourth integrates AI matching and reporting. All four members contribute to testing and combining the modules.

3) Feasibility

- With these tools and skills, the project is possible within the proposed timeline .
- Open-source tools can reduce licensing costs
- Team members already have the basic skills needed, and extra parts (like AI) can be learned during the project.

As real asset data is unavailable, development and evaluation will use sample or simulated records. Testing will follow the Phase 1 success criteria, including the planned workload of 100 asset records and five simultaneous users. AI accuracy and response-time targets remain to be demonstrated.

Overall, ReSource appears technically feasible as a prototype, provided the team confirms the required skills, validates the selected AI tools, and maintains the agreed scope.

Technical Ownership

The IT Department would be the technical owner of ReSource, responsible for hosting, technical maintenance, security updates, backups, and support in a future university deployment. During the academic prototype, the student team would manage the development and demonstration environment.

Data & AI

Data Preparation and Quality

The sample dataset described above will include asset descriptions, categories, specifications, quantities, condition, location, and availability, together with related request and maintenance records. The team will check missing fields, duplicate identifiers, inconsistent categories, and invalid quantities before using these records for matching or reporting. Missing information will be flagged rather than treated as verified data.

The prototype will use synthetic campus asset records generated with AI assistance and based on clearly stated assumptions. These records will represent assets such as furniture, computers, and laboratory equipment, with realistic descriptions, quantities, conditions, and locations. The team will review the records for completeness and consistency and verify expected categories and suitable matches before evaluation. The dataset will support development and testing and will not represent actual university inventory.

AI Contribution and Team Contribution

Pretrained AI components would provide semantic comparison, category suggestions, and generated summaries. The team's contribution will be to prepare the data, integrate these components into ReSource, define matching and approval rules, and evaluate the results. Classification suggestions will require user review, while sustainability recommendations will depend on recorded condition, maintenance information, available costs, and approved decision criteria.

The backend will enforce permissions, availability, quantities, and approvals. It will also calculate report totals, which an LLM may summarize. This separation will allow AI to support decisions while keeping operational controls and numerical calculations within the application.

The main technical and AI risks are:

- Incomplete data: Any record missing a required matching field will be flagged for correction before matching.
- Inaccurate recommendations: Results below the matching or classification targets will trigger review of the data and selected model. Manual category selection and structured search will remain available.
- Slow or unavailable AI: At least 90% of at least 20 AI interactions should return usable results within 15 seconds. Timeouts or service failures will provide a clear fallback to ordinary forms, structured search, or conventional reports.

Financial Analysis

Overview and Continuity from Phase 1

This section evaluates the Economic dimension of the TELOS feasibility framework for ReSource. It builds directly on the Expected Benefits and Assumptions stated in the Phase 1 report [8], where the team identified the primary financial benefit as “reduced spending by reusing surplus assets and lowering storage costs,” and listed as a working assumption that “the university would be willing to adopt a centralized platform for internal asset exchange if one were available.” The analysis below turns those qualitative statements into quantified, formula-backed estimates.

Consistent with the Phase 1 Project Constraints (academic prototype scope, one Fall 2026 semester, a four-student team, and reliance on existing libraries, pretrained models, or APIs rather than custom-trained models), all figures in this section describe a single-semester pilot deployment, not a full university-wide rollout. Every number that is not drawn from a cited public source is explicitly labeled as an assumption, together with the reasoning behind it, so the figure can be checked and adjusted rather than taken on faith.

Consistent with standard engineering economic-analysis practice, this section treats engineering labor as the primary, market-priced cost driver of the project rather than as a free resource. Personnel time is typically the dominant cost in a software initiative, and pricing it at a real market rate gives a realistic picture of what building and operating ReSource actually costs. The current development team's labor carries no cash outlay at this stage, which is noted separately for transparency, but that is not the basis for the economic case below.

Two categories of cost are analyzed: nonrecurring (one-time) costs and recurring (ongoing operational) costs, followed by a benefit analysis and a combined cost-benefit summary.

Nonrecurring (One-Time) Costs: Engineering Labor

Industry cost estimation for software projects is dominated by personnel cost, not infrastructure or tooling, so this section leads with a market-priced labor estimate rather than an AED 0 in-kind figure. All cost and benefit figures in this section are presented in AED (UAE dirham).

$$\textit{Professional Labor Cost} = \textit{Team Size} \times \textit{Avg. Hours/Week} \times \textit{Development Weeks} \times \textit{Market Hourly Rate} \quad (1)$$

Cited market rate: PayScale [9] reports the average entry-level Software Engineer salary in the UAE as AED 95,779/year (2026 data). Converting to an hourly rate using a standard 2,080-hour work year (40 hours/week × 52 weeks) gives AED 46.05/hour.

Assumption (hours): 4 students × 8 hours/week × 12 development weeks (spanning the design-through-development window described by Phases 2–6 of the Phase 1 timeline) = 384 person-hours. Eight hours/week per student is a reasonable load for a project layered on top of a full course schedule.

$$\textit{Professional Labor Cost} = 384 \text{ hours} \times \text{AED } 46.05/\text{hour} \approx \text{AED } 17{,}683 \quad (2)$$

Actual cash outlay at the current development stage

Because ReSource is currently being built by an academic project team rather than a commercially staffed engineering team, the real out-of-pocket labor cost at this stage is AED 0: no one is actually paid the figure above (see Phase 1 Constraint: “team of 4 students”). The AED 17,683 professional-labor estimate is still carried forward as the primary cost figure in the Cost-Benefit Summary below, since a credible economic case for ReSource should hold up under realistic market costing rather than rest on the fact that the current team's labor happens to be free.

Other nonrecurring items

Item	Real Cash Cost	Notes
Development tools (IDE, Git/GitHub, UML tools)	AED 0	Free tiers / open-source, per Phase 1 Technical Resources
Optional custom domain name	~AED 44/year	Only if the prototype is made publicly reachable; not required for course demonstration, which can run on the platform’s default subdomain

Recurring (Operational) Costs

For costing purposes, this section assumes a representative technology stack consistent with the Phase 1 Technical Resources table, which deliberately left the “Database System” and “AI Models / APIs” rows generic: Firebase/Firestore for the backend database and hosting layer, and the Claude API (Haiku 4.5 model) for the AI Matching Engine’s semantic-similarity and classification calls. This is a costing assumption only (the actual technology selection remains a Phase 4/5 design decision), but it is grounded in real, publicly published pricing so the estimate is checkable.

Database and Hosting Costs

Firebase’s no-cost Spark plan includes, per the official Firebase pricing page [10]: 1 GiB of Firestore storage, 50,000 document reads/day, 20,000 document writes/day, 20,000 document deletes/day, and 10 GiB of network egress/month, plus 10 GB of Hosting storage and 360 MB/day of Hosting data transfer. The pay-as-you-go Blaze plan keeps all of these quotas free and only bills usage above them at standard Google Cloud rates.

Assumption: a one-semester pilot with roughly 150 active users (students, department representatives, and the Procurement Team / University Management stakeholders identified in the Phase 1 stakeholder table) generates an estimated 5,000 document reads/day and 500 document writes/day, both comfortably inside the Spark free-tier quotas of 50,000 reads/day and 20,000 writes/day.

Conclusion: database and hosting costs for the pilot are AED 0/month, since projected usage stays within Firebase’s published free-tier limits. If ReSource were scaled beyond the course prototype to a university-wide production deployment (outside the Phase 1 Constraints, which explicitly scope the project as “an academic prototype… rather than a production-ready,

university-wide deployment"), usage above these quotas would bill at standard Google Cloud rates, which is flagged here as a scaling dependency rather than estimated further.

AI API Costs (Claude Haiku 4.5)

Anthropic's published API pricing [11] lists Claude Haiku 4.5 at $1 per million input tokens and $5 per million output tokens (Anthropic publishes its pricing in USD). Converted to AED at the UAE dirham's fixed peg of 3.6725 AED/USD, this is AED 3.6725 per million input tokens and AED 18.3625 per million output tokens, the most cost-effective model tier, suited to the short, structured prompts a matching or classification call requires. All calculations below use the AED-converted rate.

Cost per AI call = (Input Tokens × AED 3.6725 / 1,000,000) + (Output Tokens × AED 18.3625 / 1,000,000) (3)

Assumption: each AI Matching Engine call (comparing one request against one available asset, per the Phase 1 description: "ranks open requests against available assets using semantic similarity plus condition, location") uses approximately 500 input tokens (the two item descriptions plus a short instruction) and 150 output tokens (a compatibility score and a short explanation, as required by the Phase 1 requirement that matches include "a clear explanation").

Cost per call = (500 × AED 3.6725/1,000,000) + (150 × AED 18.3625/1,000,000) = AED 0.0018 + AED 0.0028 = AED 0.0046 (4)

Assumption: using the same 150-active-user pilot base introduced above under Database and Hosting Costs, and an average of 2 matching requests per active user per week, over a 16-week Khalifa University semester: 150 × 2 = 300 requests/week × 16 weeks = 4,800 AI calls for the semester (an average of 1,200 calls/month).

Semester AI cost = 4,800 calls × AED 0.0046/call ≈ AED 22.04 (≈ AED 5.51/month) (5)

Total Recurring Cost Summary

Cost Item	Basis	Semester Cost (16 weeks)
Firestore database & Hosting	Within Spark free-tier quotas (cited)	AED 0.00
Claude Haiku 4.5 API calls	4,800 calls × AED 0.0046/call (cited USD pricing, converted at the 3.6725 peg)	AED 22.04
Total recurring cost		AED 22.04

Benefit Analysis: Avoided-Purchase and Storage Savings

The Phase 1 Expected Benefits section states the financial benefit qualitatively: "reduced spending by reusing surplus assets and lowering storage costs." This section quantifies that

benefit directly, using the avoided purchase price of reused assets together with clearly stated assumptions so every figure can be checked and adjusted. The formula and assumptions below do exactly that, cross-checked where possible against a simulated stakeholder interview (Appendix A).

Avoided-Purchase Savings = Number of Assets Reused × Average Avoided Purchase Price (6)

Assumption (reuse volume): using the same 150-active-user pilot base introduced above under Recurring (Operational) Costs, the team assumes roughly 1 completed reuse transaction for every 4 active users over the semester (150 ÷ 4 ≈ 37, rounded to 40 assets). This is a conservative, illustrative figure for an early-stage internal exchange pilot, not a KU-specific historical statistic, since itemized KU reuse or purchase records are not publicly available. A simulated University Management (Customer) Representative (AI Stakeholder Mimic; see Appendix A, Q1) described a realistic first-semester pilot as "a handful of departments… in the range of a hundred to a couple hundred active participants," which is consistent with the 150-user scale used here.

Assumption (avoided unit price): AED 551 average avoided purchase price per reused asset, a blended estimate across the low-to-mid-value campus items the Phase 1 report describes ReSource handling (furniture, lab equipment accessories, electronics peripherals, and office supplies). The same simulated University Management Representative (Appendix A, Q2) described the assets most likely to go through the system as "the everyday stuff departments accumulate… individually low-to-mid value items, not major capital equipment," which supports keeping this estimate in the low-to-mid range rather than assuming higher-value equipment.

Avoided-Purchase Savings = 40 assets × AED 551 = AED 22,040 (per pilot semester) (7)

Assumption (storage/handling savings): AED 18 per asset in avoided storage and handling cost, for items that are matched and transferred instead of sitting in surplus storage awaiting a manual disposal decision, directly reflecting the Phase 1 benefit statement "lowering storage costs."

Storage Savings = 40 assets × AED 18 = AED 720 (per pilot semester) (8)

Total estimated semester benefit = AED 22,040 + AED 720 = AED 22,760

These figures are illustrative projections built from explicitly stated assumptions, intended to demonstrate the avoided-purchase savings mechanism described in Phase 1's Expected Benefits section. They are not confirmed Khalifa University financial data, since itemized figures of that kind are not publicly available to the team.

Cost-Benefit Summary

Two views are given below. The professional-costing view is presented first and is the primary conclusion of this analysis, since it reflects what ReSource would cost under realistic market

conditions; the actual-cash-outlay view is kept only as a secondary transparency note, since it reflects the current development stage's invoice, not the project's underlying economic value.

View	Total Cost (Semester)	Total Benefit (Semester)	Net Result
Primary: professional costing (labor priced at market rate)	AED 17,705	AED 22,760	+AED 5,055
Secondary (transparency only): actual cash outlay at the current development stage	AED 22.04	AED 22,760	+AED 22,738

Primary result: treating engineering time at real UAE market rates (AED 17,683 in professional labor, from Nonrecurring (One-Time) Costs above, plus AED 22.04 in recurring cloud/AI cost, from Recurring (Operational) Costs above), ReSource still shows a positive net result of roughly AED 5,055 for a single pilot semester, a benefit-to-cost ratio of about 1.3:1. That is a far more conservative and credible figure than simply noting that the current team's labor is free, and it is the figure University Management should rely on, since it holds up under the same costing standard a fully staffed engineering team would be measured against.

The secondary, actual-cash-outlay view is retained for completeness: at the current development stage, the real cash spent this semester is about AED 22, which is a true statement about the project's present cost but not, on its own, a meaningful measure of the system's economic value.

Assumptions Used in This Section

- Pilot scale: approximately 150 active users (students, department representatives, Procurement Team, and University Management) during a single Fall 2026 semester pilot, consistent with the Phase 1 Project Constraints scoping ReSource as an academic prototype, and consistent with a simulated University Management stakeholder interview (AI Stakeholder Mimic; Appendix A).
- Usage intensity: an average of 2 AI matching requests per active user per week, and roughly 1 completed reuse transaction per 4 active users over the semester.
- Technology stack for costing purposes only: Firebase/Firestore (database and hosting) and the Claude API, Haiku 4.5 model (AI matching and classification), consistent with, but not more specific than, the generic “Database System” and “AI Models / APIs” line items already listed in the Phase 1 Technical Resources table. Final technology selection remains a Phase 4/5 design decision.
- AI call size: approximately 500 input tokens and 150 output tokens per matching/classification call.
- Average avoided purchase price of AED 551 and avoided storage/handling cost of AED 18 per successfully reused asset, blended across the low-to-mid-value item types (furniture, lab accessories, electronics peripherals, office supplies) described in the Phase 1 report.
- Professional labor priced at the UAE market average entry-level Software Engineer rate: AED 95,779/year (PayScale, 2026), converted to AED 46.05/hour using a

standard 2,080-hour work year, applied to 384 person-hours (4 students × 8 hours/week × 12 development weeks).

- All figures in this section are presented in AED. Anthropic's Claude API pricing is published in USD and was converted using the UAE dirham's fixed peg of 3.6725 AED per USD; all other figures are AED-native (PayScale) or team-stated assumptions expressed directly in AED.

Legal and regulatory analysis

Overview and Scope

This section covers the Legal dimension of the TELOS framework in full: legal, ethical, privacy, security, procurement, ownership, and safety constraints, together with the university-policy context ReSource would operate under, not only the intellectual-property and data-privacy angle shown on the Phase 2 slide deck. It is grounded in three UAE federal laws and two real, currently published Khalifa University web pages, cited below, and it builds directly on the Phase 1 report's [8] stakeholder roles, AI Matching Engine description, and Project Constraints.

As with the Financial (Economic) Feasibility section above, every legal point is tied to a cited primary or secondary source; where no public Khalifa University policy document could be found to cite, that gap is either filled using a simulated stakeholder interview (documented in Appendix A) or stated openly as an unresolved open item under University Policy Dependencies and Open Items below rather than filled with an invented citation.

Data Privacy and Protection: UAE PDPL (Federal Decree-Law No. 45 of 2021)

ReSource stores personal data belonging to Khalifa University students and staff, including names, university IDs, department affiliations, and asset request/transfer history, which brings the system within scope of the UAE's Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data (PDPL) [12], [13], the UAE's general federal data-protection law.

- Consent (Article 6): the data controller must be able to prove the data subject's consent where consent is the lawful basis for processing, and users may withdraw consent at any time. ReSource's account-registration flow needs an explicit, clear consent step before storing user or asset data.
- Data subject rights: the PDPL grants individuals the right to access their data, request correction or erasure, restrict processing, and, most relevant to the AI Matching Engine, the right to object to automated decision-making that has a legal or similarly significant effect on them.
- Right to object to automated decisions: because the AI Matching Engine assigns a compatibility score and recommendation for each request, a user who disputes or is negatively affected by a match should be able to request a human review, not just an automated explanation.

- Data Protection Impact Assessment (Article 21): a DPIA is required before processing that involves systematic, large-scale automated evaluation with legal or significant effect. Because the AI Matching Engine performs exactly this kind of systematic automated evaluation, the team should treat producing a DPIA as a Phase 3/4 deliverable-adjacent task, not an afterthought.
- Penalties: the PDPL itself does not fix specific fine amounts; it allows the UAE Cabinet to set administrative penalties through Executive Regulations. This is noted here as a compliance risk to plan for, rather than a number to quantify, since no public Executive Regulation fine schedule could be cited.

Intellectual Property and AI-Output Ownership: UAE Copyright Law (Federal Decree-Law No. 38 of 2021)

ReSource generates AI-authored content: the Matching Engine's compatibility explanations and the "Generative AI Reporting" (financial savings, waste diversion, and avoided-purchase summaries) noted in the Phase 1 stakeholder table for University Management. UAE Federal Decree-Law No. 38 of 2021 on Copyright and Neighboring Rights [14] protects works that show originality and human creativity, and, consistent with the position most jurisdictions currently take [15], does not recognize an AI system itself as an "author."

- Practical effect: content generated by the AI Matching Engine or the reporting feature with little human involvement sits in a legal gray area, since a system cannot be classified as the natural-person author copyright protection normally requires.
- Recommended mitigation: document that KU (as the commissioning institution) owns system-generated reports and matching explanations through the system's terms of use, and log sufficient human review/curation of AI outputs (e.g., a staff member approving a generated sustainability report before it is shared) to support that ownership claim, the approach current legal commentary [15] recommends where the law itself is silent.
- Codebase ownership: who owns the ReSource source code itself at this development stage is a separate, open question the Phase 1 report does not address and that is outside the scope of this feasibility analysis; it is listed as an open item under University Policy Dependencies and Open Items below for Khalifa University's relevant office to confirm.

Cybersecurity and Unauthorized Access: UAE Cybercrime Law (Federal Decree-Law No. 34 of 2021)

ReSource is, functionally, a database of university user and asset records, which places it squarely within the scope of Federal Decree-Law No. 34 of 2021 on Countering Rumors and Cybercrimes [16]. The law criminalizes unauthorized access to IT systems and sensitive data, with penalties of 6 months to 5 years' imprisonment and fines ranging from AED 20,000 to AED 3,000,000 depending on the severity of the offense [17]; it also requires clear consent before personal data is collected, transmitted, or processed.

- Implication for design: role-based access control should follow the stakeholder roles already defined in the Phase 1 stakeholder table (e.g., the Procurement Team can

view inventory; University Management has final budget/adoption authority), so that access is scoped to what each role legitimately needs.

- Recommended controls: authentication, audit logging of who viewed or changed asset and transfer records, and encryption of data in transit and at rest, all standard measures that directly reduce exposure under this law's unauthorized-access provisions.
- KU single sign-on (SSO): per a simulated interview with an IT Department Representative (AI Stakeholder Mimic; see Appendix A, Q2), SSO integration is possible in principle for an approved internal tool but requires a formal KU IT request and review, not something a course prototype should assume by default. The design should treat standalone authentication as the working assumption for the prototype, and SSO integration as a Phase 3/4 dependency to raise with KU IT.

Procurement, Institutional Ownership, and Sustainability Alignment (Khalifa University Policy)

Two real, currently published Khalifa University web pages give the project direct institutional grounding, beyond the three federal laws above.

- KU's Procurement & Contracts Department [18] states that "procurement is more than just buying things; it's about making thoughtful choices that help our planet and future generations," and describes sustainability-linked supplier engagement and a formal Procurement Policy document. ReSource operationalizes exactly this principle (reduce new purchases by matching internal surplus first) at the department level, which is a strong feasibility argument, but it also means the tool intersects with real budget-authorization and procurement rules. The team should formally involve this department during Phase 3/4 review, since any system that influences whether a department buys new or reuses existing assets touches procurement policy and University Management's budget authority, as reflected in the Phase 1 stakeholder table. This is consistent with a simulated University Management Representative (AI Stakeholder Mimic; Appendix A, Q3), who noted that final buy-new-vs-reuse decisions still go through the department's budget holder and, above a routine amount, through Procurement, meaning ReSource should surface and support the reuse case, not make the final spending decision itself.
- KU's Sustainability Initiative [19] already runs a "Recycle, Reduce & Reuse" program, reports a 3% reduction in energy consumption from 2022, and aligns its work with the UAE Water Security Strategy 2036 and UAE Energy Strategy 2050. ReSource extends these existing, university-endorsed commitments into an AI-assisted exchange tool, which gives the project institutional legitimacy and a natural internal sponsor, the KU Sustainability Office, in addition to the IT department.

Ethical and Safety Considerations

- Matching fairness: the AI Matching Engine's ranking should not systematically disadvantage smaller departments with fewer registered assets; matching criteria and the "clear explanation" the Phase 1 report requires for each compatibility score should be documented and reviewable, so bias can actually be checked rather than assumed away.

- Physical safety in the asset life cycle: the Phase 1 report's repair/refurbish/donate/recycle/retire life cycle covers general surplus items, but some assets (lab chemicals, batteries, equipment with electrical or sharp-edge hazards) need safe-handling and proper e-waste disposal, not just a software match. Hazardous-material items should be flagged for manual safety review rather than auto-matched, to avoid a safety liability gap the current workflow does not explicitly close.
- Data minimization: only asset and request data actually needed for matching should be collected, in line with the PDPL principles discussed above under Data Privacy and Protection, rather than defaulting to collecting everything a form could capture.

University Policy Dependencies and Open Items

In the interest of transparency, this subsection states plainly what could and could not be confirmed from public sources, and what was instead informed by a simulated stakeholder interview. A dedicated, publicly available Khalifa University IT acceptable-use, data-governance, or asset-disposal policy document could not be located during this research; only the Procurement & Contracts Department and Sustainability Initiative pages cited above under Procurement, Institutional Ownership, and Sustainability Alignment are directly citable public KU sources.

Stated assumption, informed by a simulated interview: the team assumes KU's Office of Information Technology maintains an internal data-governance and account/access policy that would apply to any system handling KU user accounts. This is consistent with standard university IT governance practice and the PDPL obligations described above under Data Privacy and Protection, and it is also what a simulated IT Department Representative (AI Stakeholder Mimic; see Appendix A, Q1) described when asked directly. Because that interview is simulated, not a real statement from KU IT, this remains an assumption, and the team should still request and review the actual policy directly from KU IT during Phase 3 rather than treating it as confirmed.

Open item, partially informed by a simulated interview: how ReSource's life-cycle "retire" action for hazardous or regulated assets (electronics, lab items) should route to Facilities Management or an approved recycling vendor is not documented in any public KU source. A simulated IT Department Representative (AI Stakeholder Mimic; see Appendix A, Q3) suggested such items should route to Facilities Management or a vendor rather than being closed out inside the app, which is consistent with the recommendation above under Ethical and Safety Considerations to flag hazardous items for manual review. This should be confirmed with the real department during Phase 3.

Open item (unresolved): ownership of the ReSource codebase and of AI-generated report content at this development stage (see Intellectual Property and AI-Output Ownership above) should be confirmed with Khalifa University's relevant office (e.g., Office of the Provost or a Technology Transfer function), since no public KU IP policy for early-stage internal projects could be cited here and this was not covered by the simulated interviews in Appendix.

Legal and Regulatory Risk Summary

Risk	Governing Law / Policy	Mitigation
Processing personal data without valid consent or a lawful basis	UAE PDPL (Federal Decree-Law No. 45/2021), Art. 6	Explicit consent step at registration; document lawful basis for each data use
No mechanism to contest an AI matching decision	UAE PDPL, right to object to automated decision-making	Provide a “request human review” option on AI matches
Unclear ownership of AI-generated reports/explanations	UAE Copyright Law (Federal Decree-Law No. 38/2021)	Ownership clause in terms of use; log human review of AI outputs
Unauthorized access to user or asset records	UAE Cybercrime Law (Federal Decree-Law No. 34/2021)	Role-based access control, authentication, audit logging, encryption
Matching or disposal workflow conflicts with department procurement/budget rules	KU Procurement & Contracts Department policy	Involve Procurement & Contracts Dept. in Phase 3/4 review
Hazardous items routed through an automatic, non-specialist workflow	General safety / duty of care (institutional, not a specific cited law)	Flag hazardous-material assets for manual safety review, not auto-matching
No publicly confirmed KU IT data-governance policy for the system	Assumption, informed by AI Stakeholder Mimic; see Appendix A	Request and review KU IT’s actual internal policy directly during Phase 3

Operational Feasibility

Overview and Scope

This section evaluates the Operational dimension of the TELOS feasibility framework for ReSource: whether the departments, asset custodians, administrators, maintenance staff, procurement staff, and sustainability officers identified in the Phase 1 stakeholder table can realistically adopt the workflows ReSource proposes. Consistent with the Financial (Economic) Feasibility and Legal, Regulatory and Ethical Feasibility sections above, this assessment is scoped to a single Fall 2026 semester pilot with roughly 150 active users, not a full university-wide rollout, and every judgment below is stated as an assumption or a direct consequence of the stakeholder roles the team has already defined, rather than as confirmed organizational fact.

Current-State to Future-State Workflow Change

The Background and Context section above describes the current process as informal and undocumented: surplus assets are stored, reported for disposal, or mentioned to other departments by email or word of mouth; requests are raised without checking existing surplus; transfer approvals happen through email or paper forms with no standard workflow; and maintenance records live in separate department spreadsheets or paper logs. Operational feasibility depends on how large a jump ReSource represents from that baseline for each stakeholder group, since the size of the change, not just its direction, determines whether it is adopted.

For most of the identified stakeholders, ReSource formalizes and centralizes a task they already perform informally rather than asking them to take on an entirely new responsibility: department representatives already decide what their department needs and what it can spare; administrators already approve transfers and disposals in some form; asset coordinators already track condition and location, even if only on paper. This matters operationally, because a recent large-scale meta-analysis of technology-acceptance studies finds that perceived usefulness and perceived ease of use remain the strongest predictors of whether a system is actually adopted, with no significant weakening of that effect despite decades of technological change [20], in practice, that means a system is adopted when target users judge it as both useful for their existing task and no harder to use than their current approach. Where ReSource instead asks a stakeholder to take on a genuinely new task, such as University Management reviewing AI-generated sustainability reports it did not previously receive, that is flagged separately below as a higher-effort adoption point.

Stakeholder-by-Stakeholder Adoption Assessment

The table below applies this current-state to future-state lens to each stakeholder category from the Phase 1 stakeholder table, and gives the team's judgment of adoption feasibility for the pilot scope.

Stakeholder	Current-State Process	New Responsibility Under ReSource	Adoption Feasibility (Pilot)
Administrators	Approves transfers/disposals ad hoc by email or paper form, no shared record	Reviews a structured approval queue; approves/rejects transfers and end-of-life actions in-app	High. Formalizes an existing task; main effort is learning one new interface rather than a new responsibility.
Procurement Team	Checks stock informally, if at all, before approving a purchase request	Consults AI-generated reuse suggestions and inventory visibility before approving new purchases	Moderate. Requires a behavior change (checking ReSource before purchasing), not just a new tool; needs an explicit process rule to be effective, not just system availability.

Department Representatives	Requests/offers assets informally by email or word of mouth	Submits structured requests and surplus listings through the platform	High. Direct digitization of an existing, well-understood task.
Asset Coordinators	Tracks condition, ownership, and location in spreadsheets or paper logs, per department	Registers/updates assets in a shared system; reviews AI-suggested matches before confirming	Moderate. Data-entry discipline (accurate, timely updates) is a new behavioral requirement, not only a new tool; incomplete or stale records would directly degrade the AI Matching Engine, so this group carries a disproportionate share of the adoption risk.
University Management	Receives ad hoc, summary-level updates on cost and impact, not structured reporting	Reviews AI-generated reports summarizing financial savings, waste diversion, and avoided purchases	Moderate. This is the one stakeholder group being asked to consume an output type (structured AI-generated reporting) it did not previously receive at all, so its adoption depends on the reports being clear and short enough to fit into an existing decision-making routine, not on any hands-on system use.

Developers and Lab Instructors are excluded from the table above because they are internal project roles rather than operational adopters of the finished system: their relationship to ReSource is building and evaluating it, not using it to manage assets, so their feasibility is addressed in the Technical Analysis and course-schedule sections rather than here.

Training and Onboarding Requirements

The Phase 1 report's working assumption is that users will have basic computer literacy sufficient to use a web-based system without extensive training. Operational feasibility for the pilot depends on that assumption holding for each stakeholder group above, and on the following minimum onboarding support being available:

- A short, role-specific quick-start guide or walkthrough for each of the five stakeholder categories in the table above, covering only the actions that role performs (for

example, Asset Coordinators would not need approval-queue instructions, and Administrators would not need asset-registration instructions).

- A brief, explicit process note to the Procurement Team stating that checking ReSource for reusable surplus is now expected before a new-purchase request is approved, since this is a behavior change rather than a system feature, and behavior changes do not happen automatically just because a system becomes available.
- At least one identified point of contact (reasonably, the student development team during the pilot, and the IT Department for any future production deployment, consistent with the Technical Ownership subsection above) for stakeholders to report confusion, data errors, or approval-workflow issues.

Assumption: onboarding for a 150-user, single-semester pilot can reasonably be delivered through short written guides and brief in-person or virtual walkthroughs rather than formal training sessions, given the assumed baseline computer literacy and the fact that most stakeholders are being asked to formalize an existing task rather than learn an unfamiliar one. This assumption should be revisited before any deployment beyond the pilot scope.

Organizational Support Requirements

Beyond individual users, three organizational conditions need to be in place for the workflows above to function as designed, each of which is already flagged elsewhere in this feasibility study and is repeated here because it bears directly on operational adoption, not only on legal or technical risk.

- IT support: the IT Department is named as ReSource's Technical Owner in the Stakeholders section and would need to provide at least baseline account provisioning and issue-escalation support during the pilot, consistent with the University Policy Dependencies and Open Items discussion in the Legal, Regulatory and Ethical Feasibility section above.
- A named process owner: the current stakeholder table gives Administrators approval authority and University Management budget authority, but does not name a single owner responsible for the reuse-first policy itself (for example, confirming that Procurement is actually expected to check ReSource before approving a purchase). Without a named owner, the Procurement Team adoption point above is the most likely to fail quietly.
- Procurement and budget alignment: since the Procurement, Institutional Ownership, and Sustainability Alignment subsection above notes that any tool influencing buy-new-versus-reuse decisions intersects with real procurement policy, the pilot's operational success also depends on Procurement's existing approval process formally accommodating a reuse-check step, not only on the AI Matching Engine functioning correctly.

Adoption Risks and Fallback Behavior

The project Risk Register already identifies user adoption as a project risk (R14, “Users find proposed workflows difficult to adopt”), with mitigation and contingency already defined there; this subsection does not repeat that entry, but explains the operational reasoning behind it. The two adoption points judged Moderate rather than High above, the Procurement Team's behavior change and the Asset Coordinators' data-entry discipline, are the most plausible sources of that risk materializing, since both depend on consistent human behavior rather than on the system simply being available. If either fails during the pilot, the practical effect is degraded match quality and inventory accuracy rather than a system outage, which is consistent with the deterministic-fallback approach (manual search, conventional reporting) already adopted elsewhere in this feasibility study for AI-related failures.

Operational Feasibility Verdict

For the single-semester, roughly 150-user pilot scope defined in Phase 1 and carried through the rest of this feasibility study, ReSource is judged operationally feasible. Four of the five stakeholder groups are being asked to formalize a task they already perform, which is a comparatively low adoption barrier; the two Moderate-feasibility points, Procurement checking ReSource before approving purchases and Asset Coordinators maintaining accurate records, are behavior changes rather than technical barriers, and both are addressable with an explicit process rule and a named process owner rather than additional system features. This verdict is conditional on Khalifa University's IT Department confirming baseline account and support availability for the pilot, and on a process owner being named for the reuse-first policy before the pilot begins, both of which remain open items for Phase 3.

Risk assessment

Methodological basis. The risk assessment was structured using established risk-management guidance. NIST SP 800-30 describes risk assessment as a process that includes preparing for the assessment, conducting the assessment, and maintaining the assessment over time [21]. ISO 31000 provides a broader organizational framework built around identifying, analyzing, evaluating, treating, monitoring, and communicating risk [22]. These sources were used to shape the process, while the numerical ratings below are project-specific estimates made for this feasibility study rather than values supplied by NIST or ISO.

Project scoring method. For each identified risk, the team assigns a Likelihood score from 1 to 5 and an Impact score from 1 to 5. The overall score is calculated as Risk Score = Likelihood x Impact. This 1-5 scoring model and the thresholds below are an adaptation for this project so that risks can be compared consistently.

Score	Rating	Interpretation
1	Very Low	Unlikely / minimal effect
2	Low	Possible but limited effect
3	Medium	Reasonably possible / noticeable effect
4	High	Likely / serious effect
5	Very High	Very likely / severe effect

Risk Score	Priority
1-5	Low
6-10	Moderate
11-15	High
16-25	Critical

Source-to-risk mapping. AI-related risks were informed by the NIST AI Risk Management Framework, which emphasizes trustworthiness characteristics such as validity and reliability, security and resilience, accountability and transparency, explainability and interpretability, privacy enhancement, and fairness with harmful bias managed [23]. Generative-AI and LLM risks were further informed by the NIST Generative AI Profile [24] and the OWASP Top 10 for LLM Applications 2025 [25]. General web-application security risks were informed by the OWASP Top 10:2025 [26], while recommended security and privacy controls were aligned at a high level with NIST SP 800-53 Rev. 5 [27].

2. Risk Register

The following register applies the methodology above to the proposed semester prototype. Mitigation describes actions intended to reduce likelihood or impact before a risk occurs. Contingency describes the fallback action if the risk materializes.

ID	Risk	L	I	Score	Mitigation	Contingency	Basis
R1	Insufficient or poor-quality asset/request data for AI functions	4	5	20 - Critical	Define required fields early; validate completeness and consistency; clean records; use representative test data.	If suitable real data is unavailable, use a controlled synthetic/demo dataset and clearly state the limitation.	[7], [22], [23]
R2	AI classification or semantic matching produces inaccurate recommendations	4	4	16 - Critical	Test against predefined examples; track confidence; compare AI output with deterministic rules; require human review.	Use conventional category, keyword, filter, or rule-based matching when AI confidence is low or the service fails.	[22], [23]

R3	AI/LLM service or API integration fails or becomes unavailable	3	4	12 - High	Separate AI features behind clear service interfaces; test integrations early; define error handling and timeouts.	Use local/simple algorithms or mocked responses so the core prototype remains demonstrable.	[23], [24]
R4	Project scope is too large for the semester timeline	4	5	20 - Critical	Prioritize essential workflows; use incremental development; assign clear owners; review progress weekly.	Defer lower-priority enhancements while preserving a complete baseline workflow and core AI demonstration.	[7], [21]
R5	Integration problems between modules developed by different team members	3	4	12 - High	Define interfaces and shared data contracts early; integrate frequently; use code review and version control.	Temporarily freeze unstable features and integrate a simpler working implementation.	[7], [21]
R6	Unauthorized access or exposure of user, asset, financial, or departmental information	3	5	15 - High	Use role-based access control, secure authentication, least privilege, data minimization, protected storage, and audit logging.	Disable affected functions, rotate credentials, restore safe configurations, and use non-sensitive demo data until corrected.	[25], [26]
R7	Bias or unfair prioritization in AI-supported matching or recommendations	3	4	12 - High	Use transparent factors; test for systematic differences; document criteria; retain authorized human review.	Allow authorized overrides and require manual approval for consequential allocation decisions.	[22]
R8	AI recommendations cannot be adequately explained to users	3	4	12 - High	Display the main factors behind each recommendation, relevant confidence information, and decision rationale.	Route low-explainability or disputed recommendations to manual review instead of automatic action.	[22], [23]
R9	Web or application security vulnerabilities affect the prototype	3	5	15 - High	Apply secure authentication, access control, input validation, safe configuration, dependency management, and logging.	Disable compromised components, patch the issue, rotate secrets, and operate only unaffected modules until recovery.	[25], [26]
R10	Prompt injection or sensitive-information disclosure through the LLM assistant	3	5	15 - High	Limit accessible data/tools; validate inputs and outputs; separate trusted instructions from user content; avoid exposing secrets.	Disable the LLM feature or restrict it to non-sensitive demo data until controls are corrected.	[24], [23]
R11	Team member availability or uneven workload causes delays	3	4	12 - High	Assign primary and backup owners; maintain shared documentation; review workload and blockers regularly.	Redistribute critical tasks and reduce nonessential scope if resource availability declines.	[7], [21]
R12	Unexpected infrastructure, hosting, or AI-service cost	2	3	6 - Moderate	Use free/student tiers where appropriate; cap API usage; prefer low-cost or local tools during prototyping.	Switch to local/open-source or lower-cost alternatives and reduce AI calls to demonstration scale.	[7], [21]
R13	Sustainability or carbon-saving estimates are misleading because assumptions/data are weak	3	3	9 - Moderate	Document assumptions, formulas, data sources, and uncertainty; validate calculations with known examples.	Present results as estimates/ranges and suspend unsupported indicators until better data is available.	[22], [23]

R14	Users find proposed workflows difficult to adopt	2	4	8 - Moderate	Keep workflows simple; test mockups with representative users; align permissions and steps with real roles.	Simplify approval/request flows and provide guided instructions or a reduced workflow for the prototype.	[7], [21]

Interpretation of the highest scores. The two highest project-specific risks are data availability/quality (R1) and semester scope/time (R4), both scored 20. AI recommendation accuracy (R2) is scored 16. These values reflect the team's feasibility judgement after applying the stated method; they are not published ratings from the cited standards.

3. Project Schedule Feasibility

Schedule feasibility was evaluated by comparing the amount of work required with the semester milestones, team availability, integration dependencies, and the need to validate AI features before they become dependencies for the rest of the system. Risk-management guidance supports reviewing risks as conditions change rather than treating assessment as a one-time activity [21], [22].

Stage	Timing	Internal Target
Phase 2 - Feasibility	Current phase	Finalize feasibility evidence, quantified risk register, schedule assessment, and recommendations.
Phase 3 - Requirements	Next milestone	Finalize functional/non-functional requirements, actors, permissions, AI requirements, data requirements, and acceptance criteria.
Phases 4-5 - Architecture & Design	Design milestone	Define application/database/AI/security architecture, interfaces, UML/ER models, UI mockups, and fallback behavior.
Phase 6 - Development	Implementation milestone	Build baseline modules first, then integrate AI classification/matching, recommendation explanations, sustainability indicators, and reports.
Phase 7 - Testing	QA milestone	Run unit, integration, workflow, security, AI-quality, and user-acceptance tests; fix high-priority defects.
Phase 8 - Final Submission	Final milestone	Complete documentation, demonstration preparation, final integration, and presentation.

4. Schedule Risk Controls

Prioritize the baseline workflow first. Authentication, assets, requests, search, approval, transfer, and maintenance should work before advanced AI enhancements are expanded.

Prototype AI early. Run a small classification/matching proof of concept before the architecture depends on an external AI service.

Integrate continuously. Merge and test modules regularly so incompatibilities are discovered while they are still inexpensive to fix.

Track risks weekly. Update likelihood, impact, mitigation status, owners, and residual risk as evidence changes [21], [22].

Keep fallback paths. The prototype should retain deterministic search/filtering and manual review so AI failure does not block the entire system [23], [24].

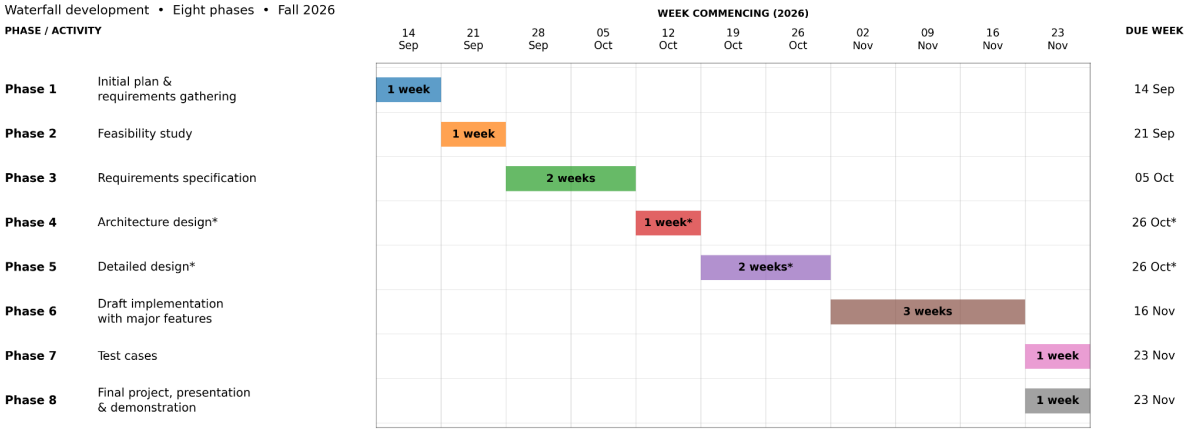
Timeline and Milestones

Phase	Main Activities	Milestone	Dependencies
Phase 1 - Project Planning and Initial Requirements	Establish the project scope, identify key stakeholders, gather preliminary requirements, recognize major constraints, and prepare an initial project plan.	Initial planning and requirements-gathering document finalized. Due: week of 14 Sep 2026	No prior dependency. This phase marks the start of the project.
Phase 2 - Feasibility Analysis	Evaluate whether the proposed system is practical from technical, financial, operational, and scheduling perspectives, including the suitability of AI technologies and the availability of required data.	Feasibility study document completed. Due: week of 21 Sep 2026	Depends on Phase 1, particularly the defined scope and preliminary requirements.
Phase 3 - Requirements Definition	Refine and document the system's functional, non-functional, AI-related, and sustainability requirements, together with business rules and acceptance criteria.	Requirements specification document completed. Due: week of 5 Oct 2026	Relies on the stakeholder needs identified in Phase 1 and the feasibility findings from Phase 2.
Phase 4 - System Architecture	Design the overall system structure, including major layers, modules, interfaces, data flows, and the planned method for integrating AI components.	Architecture section of the combined design document completed. Due: week of 26 Oct 2026	Requires the finalized requirements from Phase 3.
Phase 5 - Detailed System Design	Develop detailed database structures, workflows, UML diagrams, user-interface designs, and component-level logic for the system.	Combined architecture and detailed design document completed. Due: week of 26 Oct 2026	Depends on the completed requirements and the architecture produced in Phases 3 and 4.

Phase 6 - System Development	Implement the main asset-management functions, develop the planned AI features, and integrate the major components into a working prototype.	Draft implementation containing the main features completed. Due: week of 16 Nov 2026	Requires the architecture and detailed design produced in Phases 4 and 5.
Phase 7 - Testing and Validation	Design and carry out functional, integration, usability, security, and AI-related tests, then record results, defects, and required corrections.	Test-case documentation and testing results completed. Due: week of 23 Nov 2026	Uses Phase 3 requirements and Phase 5 design as the test basis, while Phase 6 provides the prototype to be tested.
Phase 8 - Finalization and Project Delivery	Complete final documentation, apply necessary corrections, and prepare the final presentation and system demonstration.	Final project submitted, with presentation and demonstration completed. Due: week of 23 Nov 2026	Depends on the working prototype from Phase 6 and the test outcomes and corrections from Phase 7.

ReSource | Project Gantt Chart

Waterfall development • Eight phases • Fall 2026



* Phases 4–5: proposed 1 + 2 week split. The combined design document is due in the week of 26 October.

Phases 7–8 share the final one-week window and are both due in the week of 23 November.

Source: supplied course schedule. Bars are proposed weekly allocations; exact start dates and lab-day deadlines are not specified.

Recommendations

1. Build a minimum viable baseline first. Implement stable non-AI workflows before adding advanced intelligence. This reduces schedule and integration risk.

2. Keep humans in consequential decisions. Use AI to recommend, rank, or explain; require authorized users to approve high-impact transfer, disposal, or allocation decisions. This aligns with the accountability and human-oversight orientation of the AI RMF [23].

3. Make AI behavior observable. Store recommendation inputs, scores/confidence where applicable, explanations, user decisions, and model/service version so results can be reviewed and tested [23], [24].

4. Treat the LLM assistant as a separate security boundary. Limit its tools and accessible data, validate outputs, and design specifically for prompt injection and sensitive-information disclosure risks [25].

5. Use security controls from the start. Design role-based access, authentication, logging, configuration management, incident handling, and least privilege into the prototype rather than adding them only at the end [26], [27].

6. Control scope explicitly. Classify features as essential, desirable, or future work and protect essential end-to-end functionality if schedule pressure increases [21], [22].

7. Document uncertainty. For AI and sustainability estimates, state assumptions, data limitations, and uncertainty instead of presenting estimated values as guaranteed facts [23], [24].

Constraints

Constraint	Why it matters	Simple response
Legal	The system must follow applicable laws when handling university data and AI services.	Use only approved data and services; check legal requirements before deployment.
Ethical	AI recommendations should not unfairly favor departments or make important decisions by itself.	Keep humans responsible for approvals and allow users to review or override AI suggestions.
Privacy	Asset records may include user, department, location, financial, or operational information.	Collect only needed data and limit who can view sensitive information.
Security	Unauthorized access or changes could affect asset ownership, transfers, and records.	Use authentication, role-based access, secure storage, and audit logs.
Procurement	External AI tools, cloud services, or software may require university approval or payment.	Prefer approved tools and confirm cost, licensing, and vendor requirements early.
Ownership	The system tracks who owns or controls each asset and records transfers between departments.	Require authorized approval before changing ownership, custody, or location.
Safety	Some assets may be damaged, unsafe, or unsuitable for reuse or transfer.	Require inspection or maintenance checks before risky assets are reused or transferred.
University Policy	The project must follow university rules for IT, data, assets, procurement, and disposal.	Align workflows and permissions with university procedures and responsible staff.

AI Contributions

Technical analysis was prepared by proposing the hardware and software tools, and explaining how the frontend, backend, database, and AI components would work together. The required skills were identified as well, request and transfer workflow, and outlined the prototype's constraints and planning testing workload was completed by the team members.

AI supported this work by improving wording and organization and helping draft additions on data quality, privacy, technical risks, and evaluation. The team selected relevant suggestions for inclusion and remains responsible for checking the content, confirming assumptions, and making the final decisions. It was also used as a reference to compare the available approaches to use in building the interface and the backend.

And finally AI was used to mimic stakeholders, since we have limited access to the relevant staff which are assigned to the stakeholder's roles.

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Appendix: Simulated Stakeholder Interviews (AI Stakeholder Mimic)

Due to limited direct access to the relevant Khalifa University staff within the project timeframe, the team used a structured AI-simulated stakeholder-interview technique to sanity-check its assumptions: an AI model was prompted to respond in the role of a specific stakeholder, informed by that stakeholder's publicly available institutional context (see References) and the stakeholder roles already defined in the Phase 1 stakeholder table. Its answers were used only to fill gaps that could not be resolved from public sources.

*Every answer in this appendix is clearly labeled "**A (simulated)**." It is not a real statement from any actual Khalifa University employee and is provided for planning purposes only. Wherever the body of this document cites an interview below, that citation means the assumption is informed by a simulated answer, not confirmed by it. The team should still verify these points with the real stakeholders during Phase 3 wherever possible.*

AI Stakeholder Mimic: Khalifa University IT Department Representative

Role simulated: the "IT department" Technical Owner role noted for ReSource in the team's Phase 2 feasibility notes, and referenced throughout Section 4 of this document.

Q: Does KU IT have an internal data-governance or acceptable-use policy that would apply to a new internal tool like ReSource?

A (simulated): Yes. Any system that creates or stores accounts for KU students or staff, or that touches university data, needs to go through KU IT's standard review process before it can integrate with university identity systems or handle real user data. At minimum, a new internal tool would need to align with our general data-handling and access-control expectations, which mirror what's required by the UAE's data protection law. For a course prototype like this, you wouldn't need to go through the full review to build and demo it, but you would before any real pilot with actual KU accounts.

Q: Would ReSource be allowed to use KU's single sign-on (SSO) for authentication?

A (simulated): That's possible in principle for an approved internal tool, but it would require a formal request and review, not something a course project can assume by default. For your prototype, a standalone authentication system is the safer assumption; SSO integration is something to flag as a dependency for any real deployment.

Q: Is there a specific policy for retiring or disposing of university assets (electronics, furniture, lab equipment) that ReSource's "retire" workflow should follow?

A (simulated): Asset disposal, especially electronics and anything hazardous, generally needs to go through Facilities Management and, for e-waste specifically, an approved recycling vendor, rather than being handled informally. A system like ReSource is useful for deciding whether something should be reused, donated, or retired, but the actual retirement or disposal action for regulated items should route to the appropriate department rather than being closed out inside the app itself.

AI Stakeholder Mimic: University Management (Customer) Representative

Role simulated: "University Management (Customer)" from the Phase 1 stakeholder table, described there as representing the university, confirming project priorities and budget authority, and holding final authority on budget and adoption.

Q: For a one-semester pilot of a tool like ReSource, roughly how many departments or users would realistically take part?

A (simulated): For a first pilot, I'd expect it to start small and department-level rather than campus-wide; maybe a handful of departments and their staff, plus whoever handles procurement and facilities. Something in the range of a hundred to a couple hundred active participants for a first semester sounds realistic for a controlled pilot, not thousands.

Q: What kind of assets would most likely go through a system like this, and roughly what price range?

A (simulated): Mostly the everyday stuff departments accumulate and don't need anymore: furniture, office electronics, lab accessories, sometimes small equipment. Individually these are usually low-to-mid value items, not major capital equipment. I wouldn't expect this to touch big-ticket purchases.

Q: Who would actually need to approve a final decision to buy new instead of reusing something ReSource suggests?

A (simulated): That would still go through whoever holds budget authority for that department, and for anything above a routine amount, through Procurement. ReSource can surface the reuse option and make a strong case for it, but it shouldn't be the final decision-maker on spending.